Beyond Market Data: Does Point-in-Time Public Context Add Incremental Predictive Signal?

A pre-registered test of macro, US regulatory and crypto-attention context on Binance full-asset delistings

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Abstract

We examine the premise that news, regulatory and macroeconomic context known at decision time adds predictive information to market data. Testing that premise requires two conditions that are easy to violate. Context must be reconstructed as it was available at each historical decision time, and it must be transformed into features without importing later knowledge. We test the premise in a staged design that separates the two conditions from the prediction question:

1. a data-feasibility gate asks which context layers can be reconstructed point-in-time (PIT);
2. a representation gate asks which reconstructed context can become deterministic, auditable, transform-time-PIT features;
3. a pre-registration freezes the predictive comparison before any outcome is opened;
4. the outcome is opened once.

The application is the prediction of Binance full-asset delisting announcements within 365 days. Of five candidate context layers, three were reconstructible and only conditionally so: macro, US political/regulatory, and a thin crypto-systemic layer. Exchange-specific and asset-specific context failed the data gate. Of 25 representations built from the surviving substrate, 18 passed the representation gate. No model-based text representation qualified as transform-time PIT. What survived is US macro and regulatory calendars and values plus one crypto-attention count, all common to every asset in a week, so the experiment tests whether that context predicts *when* delistings cluster, not *which* asset is delisted. The evaluation period holds about two non-overlapping 365-day label windows, and only large increments could have been detected.

In the frozen primary comparison (L2 logistic, $\lambda = 10$), adding the 18 context features to a 23-feature market baseline did not improve out-of-time precision–recall performance:

- **development:** pooled out-of-fold PR-AUC 0.2300 for the market baseline (M0) against 0.2205 with context (M2), ΔPR-AUC -0.0095. Two of eight folds were positive, and the two-way cluster-bootstrap 95% interval was $[-0.0621, +0.0405]$;

- **locked 2025 block** (not seen by any context model): 0.3270 against 0.3029, Δ -0.0240.

The development sign of Δ depends on the penalty: it is $+0.0273$ pooled under the strongest penalty in the frozen exploratory grid. The locked-block Δ is negative under every specification reported.

The pre-registered verdict is **PRE-REGISTERED NULL**. Under the frozen point-in-time specification, the surviving public-context feature set did not provide evidence of incremental out-of-time predictive information about Binance full-asset delisting beyond the structured market baseline. Both point estimates were negative, and the development interval included zero. The result concerns that surviving representation, not public context in general.

1. Introduction

Market data are the default substrate of quantitative finance. A recurring proposal is that public context, such as macroeconomic releases, regulatory actions and crypto-market news, adds information that prices and volumes do not yet carry. Testing that proposal historically demands more than a model comparison. A feature computed for a past date must use only what was public at that date, and the procedure that turns documents into features must not use knowledge acquired later. Both requirements can fail silently. A current web page about a past day may contain later revisions. A missing source may be coded as "no event". A text representation fitted on a corpus that includes later years can encode the future in its parameters even when every input document is old.

This paper reports a full research cycle built to keep those failures out of a confirmatory test. The cycle is staged so that each stage can fail without contaminating the next:

- **H20.0** asks whether context can be reconstructed point-in-time.
- **H20.1** asks whether the reconstructed substrate can be represented without future knowledge.
- **H20.2** freezes the predictive experiment before any outcome is opened.
- **H20.3** opens the outcome once and executes the frozen comparison.

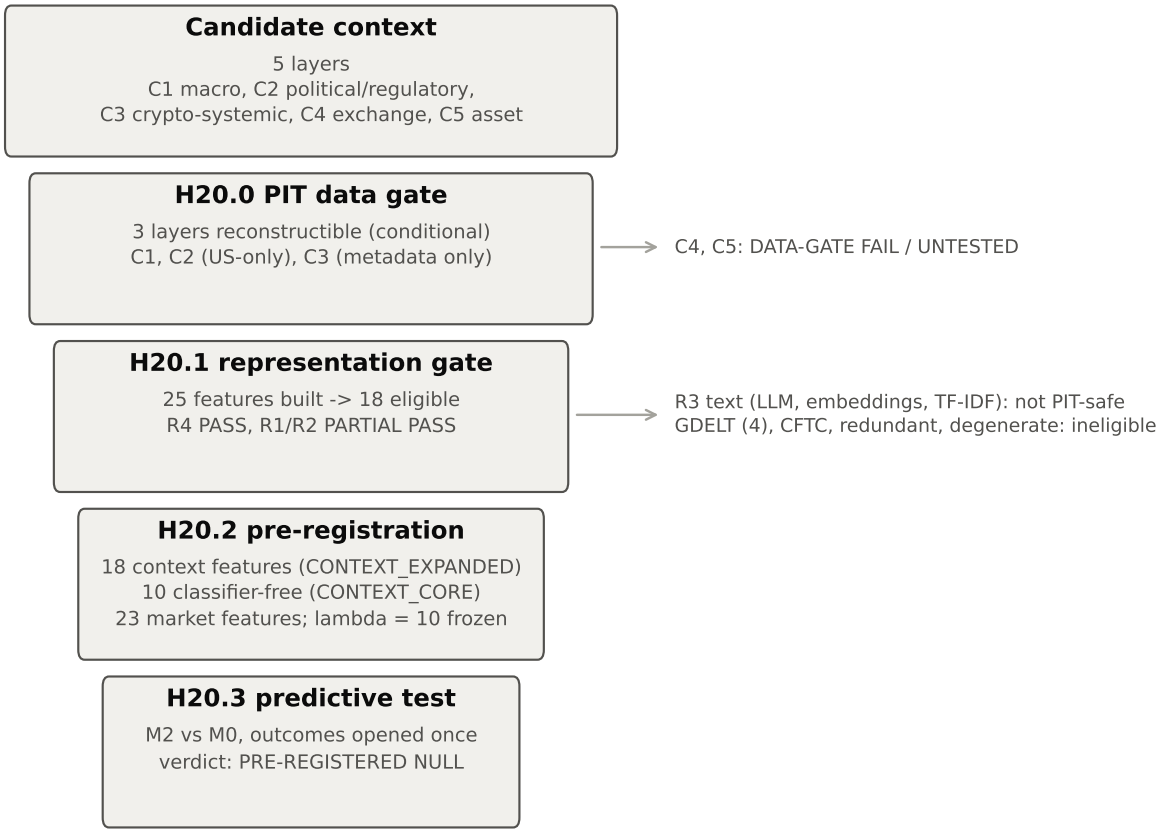


Figure 1. The Contextual Alpha research funnel. Each stage could fail independently; C4/C5 and the text representations dropped out before any outcome was opened.

The prediction task is deliberately concrete: whether a listed asset on Binance spot receives an official full-asset delisting announcement within the next 365 days. That task has a validated event universe and an established market baseline from prior work in the same repository. It also has a plausible role for context: regulatory pressure, macro conditions and crypto-market stress could precede venue decisions. Binance's current delisting FAQ lists legal/compliance, project going-concern, market and ethical risk among the factors it considers (Binance n.d.). That page is current documentation; it does not establish which criteria applied in each year of the sample.

The answer is a pre-registered null for the context that survived the gates. We report it as a result. We also report precisely what was *not* tested. The context layers arguably closest to the outcome, exchange-specific and asset-specific information, failed the point-in-time data gate and never entered the predictive experiment. That limitation is a finding of the feasibility stage, not an omission.

Relation to the companion paper. This is the second paper from the research programme that produced *Before the Backtest* (Ramos Silva 2026). The two papers share their event data and their discipline. Both use the same validated universe of 201 Binance full-asset delisting events (stage H18.1b), and both fix their design before any outcome is opened. They differ in which feasibility they check. This paper gates the data and the representation; *Before the Backtest* also asks whether a design has enough statistical support to be run at all, a check this study did not run (§7). They ask different questions. *Before the Backtest* asks whether a matched event study has enough common support to be run at all, and stops one that does not. This paper asks whether point-in-time public context adds predictive information to a market baseline. An earlier market-only test on the same delisting target, H18.3, is listed in that paper's research registry, also as a pre-registered null. This paper is a separate study: it does not revise

the companion paper, and none of its results is reported there.

2. Research question and contributions

Question. Does public context that was genuinely available at a historical decision time add incremental out-of-time predictive information beyond structured market data? In this study the question takes a narrower operational form. Every context feature that survived the gates is common to all assets in a week, and in the linear primary model such a feature moves every asset's score in that week by the same amount. The comparison can therefore gain only by ranking weeks against each other, that is, by predicting *when* delisting announcements cluster, not *which* asset is delisted (§7). The scientific object is the increment ΔPR-AUC = PR-AUC(M2) − PR-AUC(M0), not the absolute performance of either model.

Contributions. They are stated conservatively, and none is claimed as new (see §2.1):

1. **Two kinds of point-in-time validity.** We distinguish *document-time* PIT (the source record was public by t) from *transform-time* PIT (the transformation injects no knowledge unavailable at t). We operationalise the second as a classification of every candidate transform before use. The underlying risk, that a model trained on later text carries later knowledge into historical features, is established in the literature on look-ahead bias in language models (Glasserman and Lin 2023; Kelly et al. 2026).
2. **A staged design** (source reconstruction → representation audit → pre-registration → outcome opening), with closed verdict lists, frozen bars and hash-chained freezes at each stage. It applies the principle that a design is fixed before outcomes are examined (Rubin 2008; Registered Reports: Author Guidelines n.d.).
3. **An empirical case study** in which additional public context, reconstructed strictly point-in-time, did not provide evidence of incremental out-of-time predictive value over a market baseline.
4. **A documentation of the small effective sample of shared context**, and of what it implies for power. Context that is identical for every asset in a week is informed by the number of distinct weeks and of non-overlapping label windows, not by the number of asset-week rows.

2.1 Related work

Public context and crypto outcomes. The evidence is mixed and mostly concerns returns, volume or volatility over horizons of hours to days, not 365-day delisting:

- Regulatory news moves crypto prices, most strongly news about legal status, while unspecific official warnings had no measurable effect (Auer and Claessens 2018). That study measures contemporaneous market reactions, not prediction.
- Tweet counts predicted next-day trading volume and realised volatility, with weaker and mixed evidence for returns (Shen et al. 2019).
- Twitter activity and sentiment predicted short-horizon altcoin returns in a short sample, for a selected subset of coins, with low out-of-sample fit (Steinert and Herff 2018).

- The closest delisting study builds Binance delisting models whose predictors include news counts and topics and social-media metrics (Yang and Kwon 2026). Related distress and dead-coin prediction work uses market and activity variables (Będowska-Sójka et al. 2026; Fantazzini 2022; Aldhahi and Alsamaani 2026).

Look-ahead in text representations. Language models trained on unrestricted corpora can embed information from after a document's date. This look-ahead bias has been assessed for LLM sentiment scores of news headlines (Glasserman and Lin 2023). It is the motivation for point-in-time language models trained only on text available up to each date (Kelly et al. 2026). Our transform-time classification applies the same concern to every representation, including fitted statistics such as TF-IDF, before any feature is used.

Real-time information sets. Using the data vintage available at the decision date, rather than later revised data, is standard in real-time macroeconomics (Croushore and Stark 1999) and is what the ALFRED archive supports (Federal Reserve Bank of St. Louis n.d.).

Design discipline. Fixing the design before outcomes (Rubin 2008; Registered Reports: Author Guidelines n.d.) and accounting for selection across specifications (Bailey and López de Prado 2014) inform the pre-registration, as does the evidence that independent analysts given the same data and hypothesis disperse widely in their results (Menkveld et al. 2024). Survivorship and delisting bias are documented in (Shumway 1997) and (Ammann et al. 2022). Our previous paper applied an outcome-free feasibility stop to a related delisting event set (Ramos Silva 2026).

3. Why historical context is hard to reconstruct

Four failure modes motivated the design. Each was measured in H20.0 or H20.1 rather than assumed.

- **Current version ≠ historical version.** Wikipedia's Current-Events day pages are a natural historical record, but they are edited after the fact. On the H20.0 sample, 13.5% of the items present on today's pages were absent at the decision time, and 96.4% of the pages were edited after it. Reading the revision current at t avoids this; reading today's page does not.
- **Missing ≠ zero.** A source that was not reachable is not a source that reported nothing. The CFTC press-release surface returned HTTP 429 for 2021–2026 in H20.0. Coding those years as "no CFTC release" would have manufactured a regulatory lull.
- **Coverage ≠ meaning.** A keyword tag can deliver high coverage and still not mean what its name says. The inherited GDELT crypto tag reached every sampled week, but only 0.679 of its positives were crypto articles when audited (H20.1).
- **Transform-time leakage.** A 2020 document embedded by a model trained on 2026 text is a 2020 input passed through a 2026 function. Whether such a representation is a clean historical feature is a separate question from whether the document was public in 2020.

These distinctions are related to known concerns about leakage in machine-learning-based science (Kapoor and Narayanan 2023) and selection effects in backtesting (Bailey and López de Prado 2014). They are also related to the principle that observational designs should be fixed before outcome data are examined (Rubin 2008).

4. Point-in-time context framework

Layers. Context is organised by *who the information is about*:

- **C1 macro;**
- **C2 political/regulatory;**
- **C3 crypto-systemic;**
- **C4 exchange-specific;**
- **C5 asset-specific.**

Times and admissibility.

- Every record carries a public-availability time derived by fixed rules.
- Day-only documents are admitted from the next local midnight.
- Macro data vintages are admitted from the end of their vintage date in US Eastern time, so a model sees only the vintage available at t (Croushore and Stark 1999; Federal Reserve Bank of St. Louis n.d.).
- GDELT GKG records are produced in 15-minute update batches, with themes and tone assigned by GDELT's own processing (The GDELT Project 2015). As our own conservative rule, a GDELT 15-minute file is admitted 15 minutes after its stamp.
- A Wikipedia page is read at the revision current at t.
- A record is admissible at t if and only if its availability time is $\leq t$. Information at $t + 1$ minute is future information.

Document-time versus transform-time PIT. Document-time PIT is a property of inputs. Transform-time PIT is a property of the mapping from inputs to features, and has three classes:

- **PIT_SAFE:** deterministic rules whose parameters were fixed without reference to post-t documents. This covers counts, publisher categories, fixed dictionaries of generic terms, fixed arithmetic on as-of vintages, and attributes computed by a publisher at ingest time.
- **NON_PIT:** any model or statistic fitted on a corpus that includes post-t documents. For language models this is the look-ahead risk documented by (Glasserman and Lin 2023) and (Kelly et al. 2026).
- **UNRESOLVED:** plausibly safe but not demonstrated.

Only PIT_SAFE features may enter a confirmatory design.

Governance.

- Every outbound request is logged and refused if it contains outcome vocabulary.
- Raw bytes are write-once.
- Every stage freezes its plan and rules before evaluation, records hashes, and logs every departure as a deviation.
- Verdicts come from closed lists. A criterion that could not be measured is not passed.

5. Context data feasibility (H20.0)

H20.0 asked whether a sufficiently complete, timestamped, reproducible point-in-time context corpus could be reconstructed. It drew 35 quarterly decision timestamps (2018-Q1 to 2026-Q3) and registered 32 candidate sources, including rejected ones. It collected 141,751 documents for the sample windows, and tested each layer against frozen bars:

- coverage;
- year span;
- timestamp quality;
- version quality, i.e. the share of documents whose content as of t is reconstructible (bar $\geq$ 0.70);
- refetch reproducibility;
- licensing;
- entity resolution for C4 and C5 (bar $\geq$ 0.80).

Table 1. Context layers and H20.0 data-gate status

layer	G4 version share	G7 entity resolution	failed criteria	severe biases	H20.0 status
C1 Macro	0.986	n/a	none	archive_captured_le_t:S_FED_PRESS	CONDITIONAL (reconstructible)
C2 Political / regulatory	0.875	n/a	none	archive_captured_le_t:S_FED_PRESS	CONDITIONAL (reconstructible)
C3 Crypto-systemic	1.0	n/a	none	language_crypto_tag_rate; archive_captured_le_t:S_GDELT_GKG	CONDITIONAL (reconstructible)
C4 Exchange-specific	0.385	0.795	G4_version, G7_entity	archive_captured_le_t:S_BINANCE_CMS	DATA-GATE FAIL / UNTESTED
C5 Asset-specific	0.398	0.581	G4_version, G7_entity	archive_captured_le_t:S_BINANCE_CMS; entity_popularity_any_doc_w30	DATA-GATE FAIL / UNTESTED

Verdict: PARTIAL PASS.

- **C1 macro:** reconstructible, conditional on temporal and archive bias.
- **C2 political/regulatory:** reconstructible, conditional, and US-only.
- **C3 crypto-systemic:** reconstructible only in a limited form, as metadata and structured records, conditional on archive and language bias. The English crypto tag fired on 0.54% of English GDELT records against 0.14% of translingual records.
- **C4 exchange-specific:** failed. Binance notice text could be shown to read the same at t for only 0.385 of documents.
- **C5 asset-specific:** failed on version quality (0.398) and point-in-time naming (0.581).

Both C4 and C5 are **DATA-GATE FAIL / UNTESTED**, which is a statement about the data, not an economic rejection.

6. Representation feasibility (H20.1)

H20.1 asked whether the C1–C3 substrate could become stable, reproducible, auditable features without future knowledge. It froze its representation rules before drawing an independent evaluation sample of 105 monthly decision timestamps (2018-01 to 2026-09). It then built 25 features in four families:

- **R1:** event flags;
- **R2:** counts and intensities;
- **R3:** text-derived representations;
- **R4:** as-of macro values.

Transform-time audit. Current LLM extraction, current sentence embeddings and TF-IDF fitted on the full corpus were classified NON_PIT. TF-IDF refitted at each t, TF-IDF on a frozen pre-2018 corpus, and pre-2018 static word embeddings were classified UNRESOLVED. None was built. The only R3 feature built was a tone score computed by GDELT at ingest time. Point-in-time language models trained only on earlier text (Kelly et al. 2026) are a possible future route for R3; none was evaluated here.

Integrity.

- 0 point-in-time violations over 55,561 provenance records;
- byte-identical deterministic rebuilds;
- complete provenance;
- missing values never encoded as zero.

Semantic audit. An independent sample of 272 documents was audited against a frozen bar of precision ≥ 0.90. The labels were produced by the AI research agent, not by a human. That limitation is disclosed throughout, and no classifier was retuned.

- The SEC, Federal Register, enforcement, FOMC and Wikipedia classifiers met the bar.
- The GDELT crypto tag did not: precision 0.679, mostly from articles that mention crypto only in passing.
- The crypto-stress dictionary had no positives at all in the evaluation windows.

Table 2. Representation families and H20.1 eligibility

family	H20.1 verdict	built	eligible	ineligible (criterion)
R1	PARTIAL PASS	6	4	F24: REDUNDANT with F23 (rho 0.9996); F25: semantic defect K5; SPARSE_DEGENERATE (nonzero share 0.0)
R2	PARTIAL PASS	15	11	F17: coverage 0.314, 4 years; F19: semantic defect K6; F20: semantic defect K6; F21: semantic defect K6
R3	DATA-GATE FAIL / UNTESTED	1	0	F22: semantic defect K6
R4	PASS	3	3	none

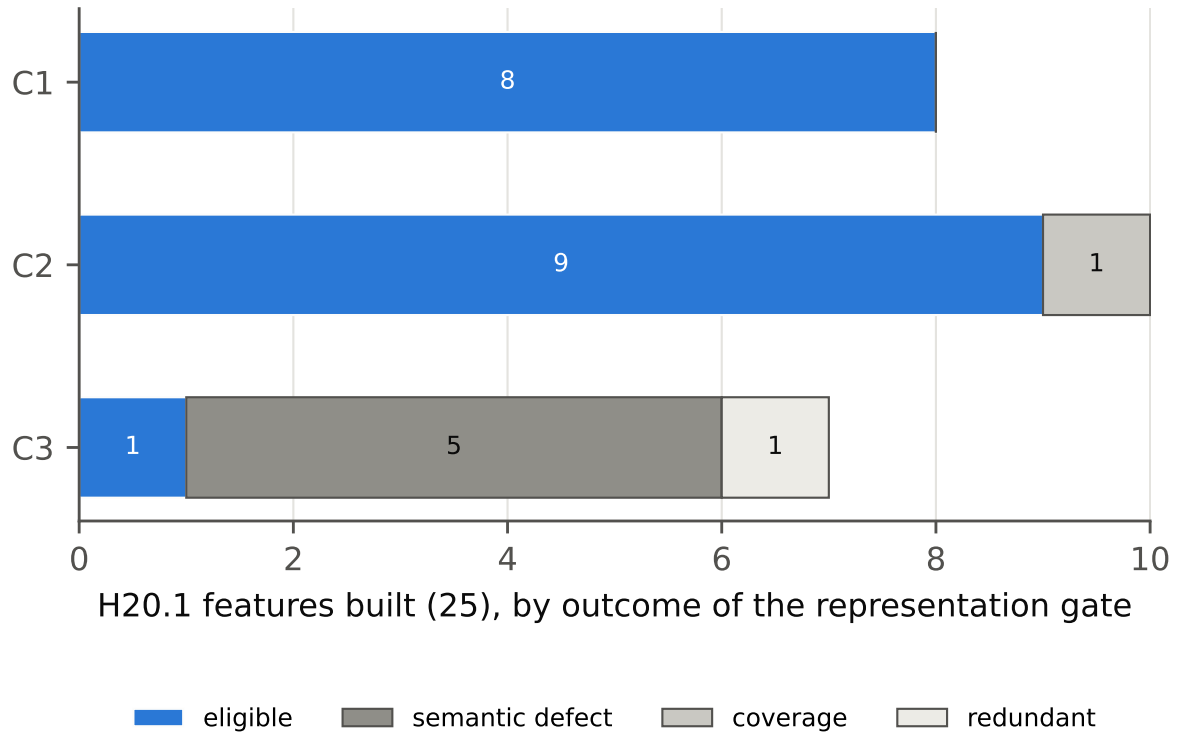


Figure 2. H20.1 feature survival by context layer. Of 25 features built, 18 were eligible; C3 lost every GDELT feature to the semantic audit.

Verdict: PARTIAL PASS.

- **R4:** PASS.
- **R1, R2:** PARTIAL PASS.
- **R3:** DATA-GATE FAIL / UNTESTED.

The 18 eligible features are:

- 8 in C1 (Fed releases and FOMC statements; CPI and unemployment release flags; the as-of policy rate, CPI inflation and unemployment);
- 9 in C2 (Federal Register financial-regulator and crypto documents; SEC releases, enforcement and crypto releases; Fed enforcement; a crypto regulatory-action flag);
- 1 in C3.

The single C3 survivor is **generic English-Wikipedia crypto attention**: a count of crypto items on the Current-Events pages. It is not a systemic-crypto-state variable.

7. Pre-registered predictive design (H20.2)

H20.2 froze the predictive experiment before any outcome was opened. It was hash-chained to the H20.1 freeze and amended once (Addendum 02), before outcomes, as follows:

- it replaced the v1 primary rule, which selected λ from $\{0.1, 1, 10, 100, 1000\}$ by mean fold PR-AUC, with a single a-priori $\lambda = 10$, and kept the v1 grid as an exploratory analysis;
- it renamed the 2025 block;
- it made the calibration criterion machine-readable.

The amendment was made before any outcome was opened. Its effect is known only now, and is reported in §9 and Table 7. Under the v1 rule both M0 and M2 select $\lambda = 1000$, the development Δ is $+0.0273$ and the LCUV Δ is -0.0167. S4 fails under either rule, so the verdict under the v1 rule would also have been PRE-REGISTERED NULL.

Unit and decision time. The unit is an asset-era × weekly landmark. The asset-era follows the inherited rule that a ticker starts a new era only after more than 365 days with no served route. The decision time is **Tuesday 00:00 UTC after the landmark Monday**, so the landmark day's daily candle is complete at t. The design is landmark-style: at each landmark, the risk set is the assets still at risk, and predictors use only information available at the landmark, as in landmarking for dynamic prediction (Putter and van Houwelingen 2017). We model a binary 365-day outcome with logistic regression; no survival model is fitted.

Target. y = 1 if and only if the asset-era's validated full-asset delisting *announcement* falls in $(t,\ t + 365$ days]. The announcement time comes from the frozen event universe and is never the trading end. Landmarks whose announcement was already public at t are excluded from the risk set.

Features.

- **M0 (23 market features):** liquidity, volatility and trailing return; route state (active pairs, quote assets); venue regime (listed assets, removal intensity, a calendar trend); and route history.
  - Inherited route-history features that rest on a class taken from Binance notice titles (C4) were excluded and replaced by notice-free versions, so the market baseline contains no venue-notice input.
  - Current listing status, whole-era quantities and every outcome column were excluded.
- **CONTEXT_EXPANDED:** the 18 H20.1-eligible features, computed at every landmark with the unchanged H20.1 rules.
- **CONTEXT_CORE:** the 10 features that use no agent-audited text classifier, fixed before any outcome as a robustness set.

Arms.

- **M0:** market only;
- **M2:** market + CONTEXT_EXPANDED;
- **M1:** context only;
- **M1_core, M2_core:** robustness arms.

The primary comparison is **M2 versus M0**.

Model. L2-penalised logistic regression with a single a-priori $\lambda = 10$, identical for every arm. The objective is Σ log-loss $+ \frac{1}{2}\lambda\|\beta\|^2$ on standardised features. Preprocessing is identical for all arms:

1. a fixed transform per feature;
2. winsorisation at the training 0.5th and 99.5th percentiles;
3. training-median imputation with missing indicators;
4. standardisation.

There is no class weighting and no recalibration. The λ grid and a LightGBM model were frozen as *exploratory* only, unable to determine the verdict.

Temporal design.

- **Development:** landmarks 2018-01-01 to 2023-12-25, with eight rolling-origin folds (Hyndman and Athanasopoulos 2021). Each fold has an expanding training window, a 365-day embargo and a quarterly test window in 2022–2023. The embargo follows from the label: a training landmark's 365-day outcome window must end before its fold's test window begins, so no delisting event can label both a training row and a test row.
- **Embargo:** a one-year embargo follows development.
- **LCUV:** a single locked block of landmarks 2025-01-06 to 2025-10-06, called the LOCKED_CONTEXT_UNSEEN_VALIDATION (LCUV) block.
  - No context or H20 model had seen it, and every rule was frozen before H20 scoring.
  - It is *not* an untouched holdout: the earlier H18.3 programme had observed market-only model behaviour in that period.
- **PROSPECTIVE_CONFIRMATION:** landmarks 2025-10-13 to 2026-09-28. It is the only block untouched by the programme and remains unopened until its outcomes mature.

What the comparison can detect. Context is common to every asset in a week. In the linear primary model it cannot reorder assets within a week, so it can change a pooled PR-AUC only by ranking weeks, which amounts to predicting the time variation of the 365-day delisting rate. The information for that is small:

- the out-of-fold evaluation spans 104 weeks;
- each week's label looks 52 weeks ahead, so consecutive labels overlap almost entirely, and the evaluation period holds about two non-overlapping label windows.

The pre-registration recorded that low power made a null the likely outcome under modest effects. It neither quantified the power nor set a stop rule, and no statistical-feasibility gate of the kind the companion paper argues for was run. In our judgement such a gate, applied before the freeze, would have flagged this comparison as underpowered and pointed the effort at the exchange- and asset-specific layers, which are the ones that could reorder assets. A post-hoc yardstick is given in §8.3.

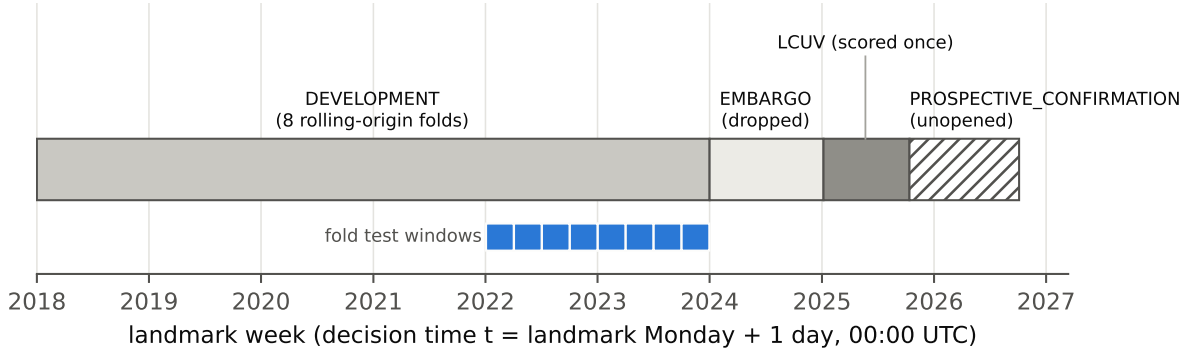


Figure 3. Temporal design. Decision time is the end of each landmark Monday; the prospective block is unopened.

Metrics and inference.

- **Primary:** PR-AUC (average precision), because the positive class is rare and ROC-based summaries can be misleading under heavy class imbalance (Saito and Rehmsmeier 2015).
- **Secondary:** ROC-AUC, Brier score, within-week top-decile lift and calibration.
- **Uncertainty:** a 2,000-replicate two-way bootstrap on the fixed out-of-fold predictions.
  - It adapts the pigeonhole bootstrap (Owen 2007), which resamples the row entities and the column entities of a crossed design independently, with replacement. Here the rows are asset-eras and the columns are the eight fold test windows (quarterly calendar blocks), not individual weeks. Each landmark row is weighted by the product of its asset-era and block multiplicities, and the statistic is a weighted PR-AUC difference.
  - Owen's consistency results concern means, and Owen describes the procedure as mildly conservative there. For a rank-based statistic such as PR-AUC they are an approximation, not a guarantee.
  - Rows are clustered because observations of one asset and of one period are correlated, and ignoring that correlation biases conventional standard errors (Petersen 2006).

Promotion rule. Support requires all of S1–S7:

- **S1:** positive development Δ;
- **S2:** at least five of eight positive folds;
- **S3:** a bootstrap lower bound above zero;
- **S4:** a positive LCUV Δ;
- **S5:** no severe calibration deterioration on the LCUV block;
- **S6:** a clean PIT audit;
- **S7:** a result not driven by one era: $\Delta > 0$ in at least two of the three era blocks (the 2022 folds, the 2023 folds and LCUV), and a development Δ that stays positive when each fold is left out in turn.

The closed verdict list is INCREMENTAL CONTEXT SIGNAL SUPPORTED, PRE-REGISTERED NULL, NEGATIVE INCREMENTAL RESULT and DATA / DESIGN FAILURE.

8. Empirical results (H20.3)

8.1 Execution and audits

H20.3 followed the frozen P0–P6 sequence.

- **Context at every landmark** (353 decision times × 18 features):
  - zero records available after t and zero post-t joins;
  - complete provenance;
  - byte-identical deterministic rebuilds.
- **Route-state fallback.** The pre-registered structural trigger fired: window-based and candle-based route states disagreed on 4.554% of landmarks, above the frozen 1% bar. The frozen candle-based fallback was therefore applied. This was a structural rule computed before any outcome, not a post-result change.
- **X sealed before the outcome.** The feature matrix was sealed and hashed before the lifecycle table was opened.
- **Target.**
  - Strict timestamp parsing found no failures, and no event matched more than one asset-era.
  - The independently rebuilt target agreed with the inherited H18 label on all 93,394 matched rows.
- **Primary LCUV scoring** ran exactly once.

The PIT, target and execution audits all passed.

8.2 Effective sample

Table 3. Temporal design and effective sample size

block	landmark Mondays	landmark rows	asset-eras	weeks	unique events	positive landmark rate
DEVELOPMENT	2018-01-01 to 2023-12-25	78227	522	313	110	5.82%
EMBARGO (dropped)	2024-01-01 to 2024-12-30	not built		53		
LCUV	2025-01-06 to 2025-10-06	15167	443	40	86	11.79%
PROSPECTIVE_ CONFIRMATION (unopened)	2025-10-13 to 2026-09-28	not built		51		

Positive landmark rows are not independent events. One delisting is the label of up to 52 consecutive weekly landmarks. Development contains 110 unique events, and the out-of-fold evaluation spans 104 distinct weeks. Because context is common to every asset in a week, its evidence rests on those weeks rather than on the 78,227 development rows.

8.3 Primary result

Table 4. Development pooled out-of-fold results (L2 logistic, $\lambda = 10$)

arm	role	PR-AUC	ROC-AUC	Brier	top-decile lift (within-week)	within-week PR-AUC	calibration slope
M0 market	primary	0.2300	0.767	0.0512	4.51	0.247	0.67
M2 market + context	primary	0.2205	0.750	0.0521	4.60	0.244	0.52
M1 context only	secondary	0.0674	0.540	0.0592	0.64	0.059	0.17
M2_core market + CONTEXT_CORE	robustness	0.2303	0.759	0.0522	4.61	0.245	0.58
M1_core CONTEXT_CORE only	robustness	0.0675	0.541	0.0573	0.64	0.059	0.20

Table 5. LOCKED_CONTEXT_UNSEEN_VALIDATION (2025) results, scored once (L2 logistic, $\lambda = 10$)

arm	role	PR-AUC	ROC-AUC	Brier	top-decile lift (within-week)	within-week PR-AUC	calibration slope
M0 market	primary	0.3270	0.809	0.0929	3.21	0.350	1.05
M2 market + context	primary	0.3029	0.806	0.0960	3.12	0.330	0.99
M1 context only	secondary	0.1193	0.506	0.1073	0.87	0.118	0.34
M2_core market + CONTEXT_CORE	robustness	0.3112	0.808	0.0972	3.10	0.337	1.01
M1_core CONTEXT_CORE only	robustness	0.1212	0.510	0.1091	0.87	0.118	1.25

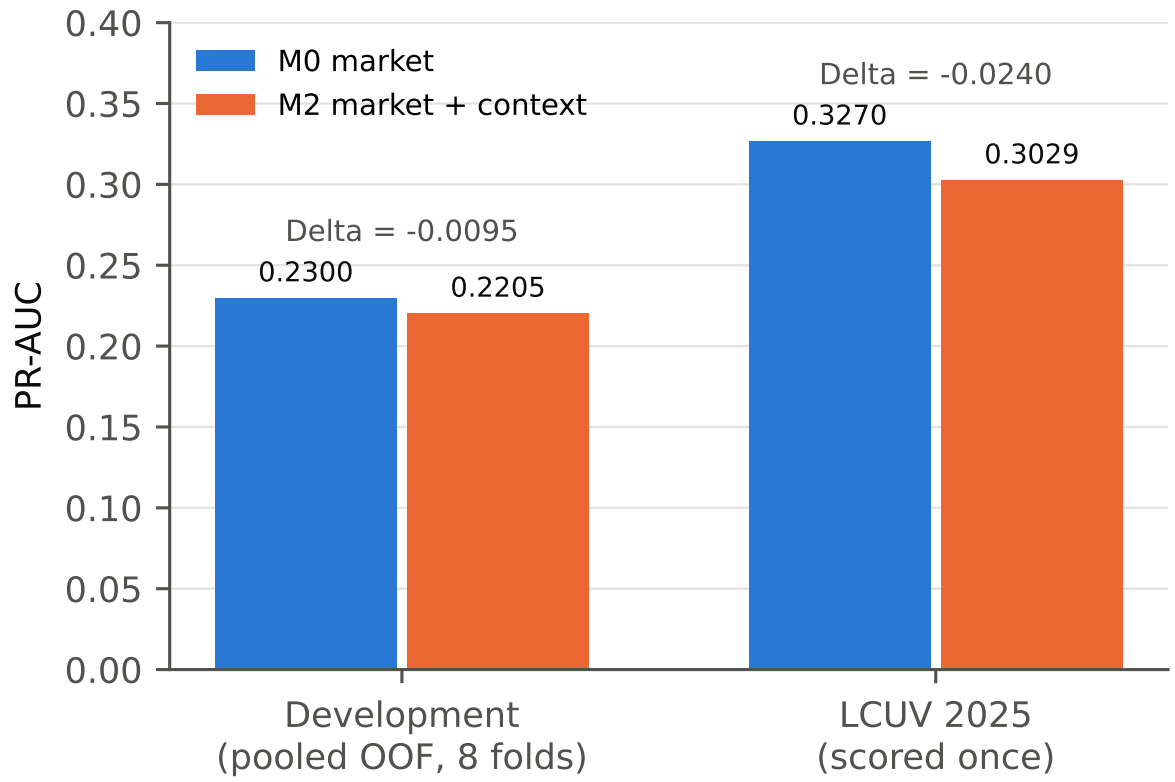


Figure 4. Primary PR-AUC, market baseline (M0) versus market + context (M2), in development (pooled out-of-fold) and on the LCUV block. Levels are not comparable across the two blocks: the positive landmark rate is 5.82% in development and 11.79% on LCUV.

Development.

- Pooled out-of-fold PR-AUC was 0.2300 for M0 and 0.2205 for M2: **ΔPR-AUC = −0.0095**, a relative change of −4.1%.
- Two of eight folds were positive.
- The two-way cluster-bootstrap 95% interval was [−0.0621, +0.0405].
- **Post-hoc yardstick (not pre-registered).** The interval's half-width implies a standard error of about 0.026. At that precision an increment would need to be about +0.07 PR-AUC, roughly 30% of M0's 0.2300, to be detected with 80% power at the 5% level. Smaller increments were unlikely to be detected.

LCUV (scored once). PR-AUC was 0.3270 for M0 and 0.3029 for M2: **ΔPR-AUC = −0.0240**, relative −7.4%. An interval from a two-way bootstrap over asset-eras and calendar months (2,000 replicates) was [−0.061, +0.001], with 3.05% of replicates positive. It is informational only: the pre-registration assigned the confirmatory interval (S3) to development.

Secondary metrics. ROC-AUC and Brier score moved in the same direction as PR-AUC. Calibration on the LCUV block did not deteriorate severely (Brier ratio 1.033 ≤ 1.05; slope excess −0.044 ≤ 0.20). In development, M2's calibration slope was lower than M0's (0.52 against 0.67). S5 does not assess this, because it is defined on LCUV; it is consistent with the added week-level features adding variance under weak shrinkage (§9).

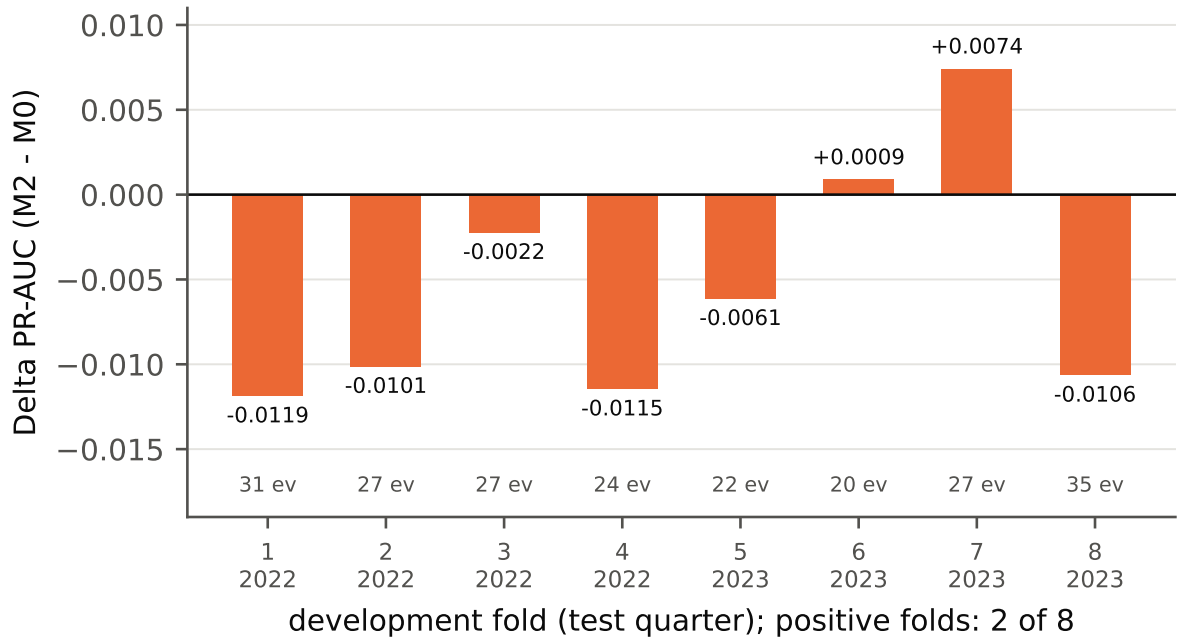


Figure 5. Fold-level ΔPR-AUC (M2 − M0) for the eight development folds, with unique test events per fold.

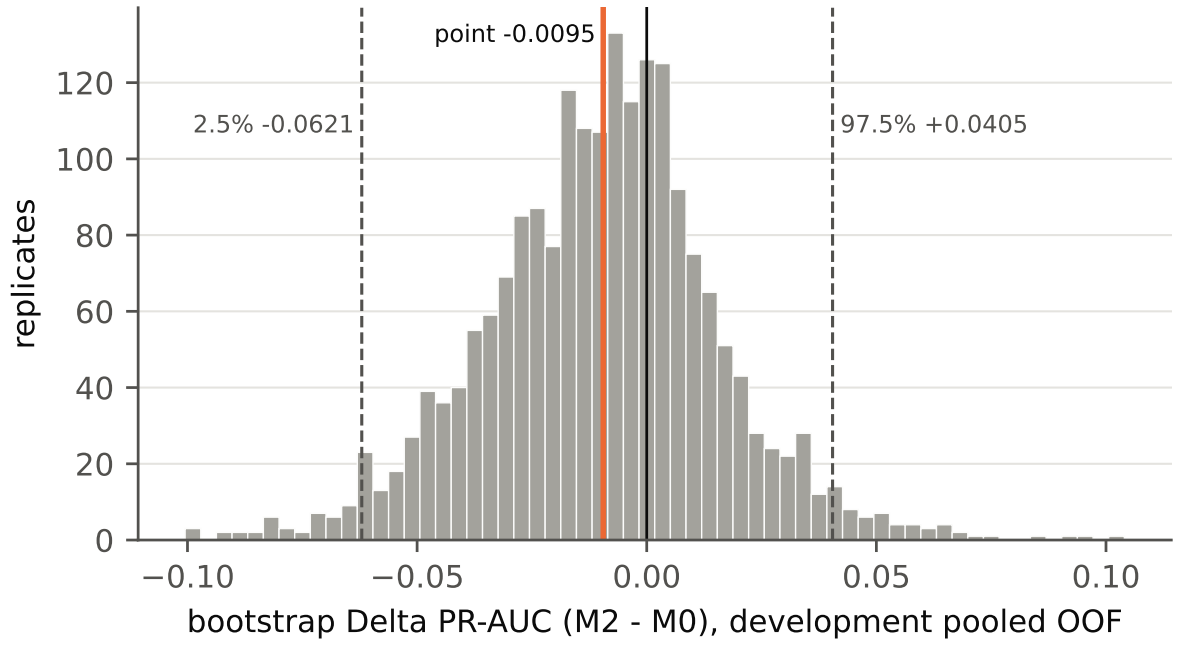


Figure 6. Two-way cluster-bootstrap distribution of the development ΔPR-AUC (2,000 replicates); dashed lines mark the 95% interval.

8.4 Verdict

Table 6. Pre-registered criteria S1–S7 and the mechanical verdict

criterion	condition	observed	result
S1	development pooled-OOF $\Delta > 0$	−0.0095	FAIL
S2	≥ 5 of 8 folds with $\Delta > 0$	2 of 8	FAIL
S3	two-way bootstrap 95% CI lower bound > 0	[−0.0621, +0.0405]	FAIL
S4	LCUV $\Delta > 0$	−0.0240	FAIL
S5	no severe calibration deterioration on LCUV (Brier ratio ≤ 1.05; slope excess ≤ 0.20)	Brier ratio 1.033; slope excess −0.044	PASS
S6	PIT audit passes	all stages PASS	PASS
S7	$\Delta > 0$ in ≥ 2 of 3 era blocks (2022 folds, 2023 folds, LCUV) and leave-one-fold-out $\Delta > 0$	1 of 3 era blocks positive; min leave-one-fold-out Δ −0.0143	FAIL
verdict	closed list, applied mechanically	PRE-REGISTERED NULL	

S5 and S6 passed. S1, S2, S3, S4 and S7 failed. No DATA / DESIGN FAILURE condition held: the PIT, target, reproducibility and execution audits passed. The result is not a NEGATIVE INCREMENTAL RESULT, because the development interval's upper bound (+0.0405) is above zero. The mechanical verdict is **PRE-REGISTERED NULL**.

9. Robustness and exploratory results

Table 7. Primary, robustness and exploratory results (only the primary determines the verdict)

analysis	status	development Δ, pooled OOF	development Δ, mean of folds	LCUV Δ	note
primary: logistic, $\lambda = 10$ (A02)	primary	−0.0095	−0.0055	−0.0240	verdict: PRE-REGISTERED NULL
CONTEXT_CORE (classifier-free), $\lambda = 10$	robustness	+0.0003	−0.0039	−0.0158	two-way CI [−0.042, +0.045]; 3 of 8 folds positive
logistic, fold-selected λ (the v1 primary rule)	exploratory	+0.0273	+0.0003	−0.0167	$\lambda = 1000$ selected for both M0 and M2; verdict under the v1 rule: PRE-REGISTERED NULL (S4 fails)
LightGBM, a-priori configuration	exploratory	+0.0285	+0.0116	−0.0080	7 leaves, 200 trees for every arm; M0 pooled PR-AUC 0.174
LightGBM, fold-selected	exploratory	+0.0271	+0.0107	−0.0030	selected: M0 15 leaves, 200 trees; M2 7 leaves, 200 trees

Classifier-free context (CONTEXT_CORE, secondary). Replacing the expanded set with the 10 features that use no agent-audited classifier gave:

- a development Δ of +0.0003 pooled (−0.0039 as a mean of folds), with a two-way interval of [−0.042, +0.045];
- an LCUV Δ of −0.0158.

The null pattern persists in the classifier-free robustness specification. This shows that the pattern is robust to removing those classifiers; it does not show that the classifiers had no influence on the expanded-set result.

Penalty path (read from the frozen exploratory run; no refit).

Table 8. Development Δ along the frozen λ grid (logistic; read from the frozen exploratory run, no refit; pooled values exist only for $\lambda = 10$ and the selected λ)

λ	M0 PR-AUC, mean of folds	M2 PR-AUC, mean of folds	Δ, mean of folds	Δ, pooled OOF
0.1	0.2242	0.2195	−0.0047	
1	0.2246	0.2199	−0.0048	
10 (primary)	0.2283	0.2228	−0.0055	−0.0095
100	0.2467	0.2411	−0.0056	
1000	0.2793	0.2796	+0.0003	+0.0273

The development Δ depends on the penalty:

- The primary $\lambda = 10$ is far from the baseline's own best. M0's mean-of-folds PR-AUC rises by about 22% from $\lambda = 10$ to $\lambda = 1000$, the edge of the frozen grid, and the optimum may lie beyond it.
- Along the grid, the mean-of-folds Δ is negative up to $\lambda = 100$ and +0.0003 at $\lambda = 1000$.
- At $\lambda = 1000$ the pooled out-of-fold Δ is +0.0273, while the mean-of-folds Δ is +0.0003. A gain that appears only when the eight folds are pooled comes from aligning score levels across quarters, not from ranking within a quarter. That is the week-level channel described in §7.
- The LCUV Δ is negative at both penalties for which it was computed: −0.0240 at $\lambda = 10$ and −0.0167 at $\lambda = 1000$.

The development sign of Δ is therefore not robust to the penalty, and the locked-block sign is negative under every specification reported. The verdict does not depend on the penalty, because S4 fails under both the A02 rule and the v1 rule.

Flexible models (exploratory; cannot change the verdict). LightGBM behaves differently from the logistic model. Its development gain is +0.0285 pooled and +0.0116 as a mean of folds, so part of it is ranking within folds, which a nonlinear model can obtain by letting week-level context interact with asset-level features. Two qualifications apply:

- The a-priori LightGBM baseline is weak. Its M0 reaches a pooled development PR-AUC of 0.174, against 0.2300 for the logistic M0, so the gain is measured from a lower starting point.
- The gain did not carry into the locked block: the LCUV Δ is -0.0080 for the a-priori configuration and -0.0030 for the fold-selected one.

Fold-selected configurations were chosen on the folds that score them, so their development values carry selection optimism. The LightGBM selection also differs between arms (15 leaves for M0, 7 for M2).

Context-only arms. Context-only models (M1, M1_core) showed little discrimination: development ROC-AUC 0.540 and 0.541. This is consistent with context ranking weeks rather than assets.

10. Discussion

The study did not support the incremental-context hypothesis. The surviving representation, built and audited as it was, did not improve out-of-time predictive performance for this target beyond this baseline. The point estimates were negative, but the development interval included zero and is wide enough that neither a negative nor a moderate positive increment is excluded. Several interpretations are compatible with this finding. None is identified by the experiment, and they are offered as hypotheses, not mechanisms.

1. **Redundancy.** Part of the public context may already be reflected in market variables such as liquidity, volatility and route structure, which respond to the same conditions.
2. **Target mismatch.** Macro and regulatory context can move market-wide risk without discriminating *which* asset will be fully delisted within a year.
3. **Cross-sectional limitation.** Context is identical for every asset at a given week, so in the linear primary model it can rank weeks but not assets (§7). The context-only arms' near-chance ROC-AUC is consistent with this.
4. **Horizon mismatch.** Short-lived public events may be weak predictors of an asset-specific decision up to 365 days later.
5. **Representation limitation.** The context most plausibly relevant to an asset's delisting, exchange-specific notices and asset-specific information, failed the point-in-time data gate and was never allowed into the confirmatory experiment. Stronger representations of those layers could behave differently.

The exploratory pattern is itself informative for practice. Tuned and more flexible models produced pooled development gains of about +0.03 that reversed sign on the locked block; for

the logistic model the gain appeared only when folds were pooled (§9). In a study without a frozen a-priori primary and a locked block, that pattern could have been reported as a context effect.

11. Limitations

- **Untested context.**
  - C4 and C5 failed the data gate and were not tested.
  - GDELT features failed the semantic audit.
  - No modern LLM, modern embedding or non-PIT text transform was used.
  - The C3 representation is thin: generic English-Wikipedia crypto attention.
- **Scope of the surviving context.** C2 is US-only. C1–C3 inherit H20.0's archive and language conditions.
- **Low power, quantified only post hoc.** The comparison ranks weeks (§7). The out-of-fold evaluation spans 104 weeks and the LCUV block 40, with 110 and 86 unique events, and the evaluation period holds about two non-overlapping 365-day label windows. Only increments of roughly +0.07 PR-AUC would have been detected reliably (§8.3). No statistical-feasibility gate was run before the freeze.
- **Penalty dependence.** The development sign of Δ changes along the frozen λ grid (§9). The verdict does not.
- **Agent-produced semantic audit.** Eight of the 18 context features depend on classifiers audited by the AI research agent. The null pattern persists in the classifier-free robustness specification.
- **The LCUV block** is context-unseen, not programme-untouched. Prospective confirmation is pending.
- **Verifiability of the freeze order.** That each freeze preceded the outcome is attested by the author's own version history and by the hashes in the freeze files. No third-party timestamp, such as a registry deposit, exists for H20.0–H20.3.
- **Market substrate approximations.** Route state is reconstructed from daily candles; the frozen fallback addressed window gaps. Two events (IDRT, VAI) have no candles and two others (SKY, USDS) attach to no era under the inherited era rule; the rebuilt target agreed with the inherited label on every matched row.
- **Single venue and target.** Results concern Binance full-asset delisting announcements and do not extend to other venues, targets or horizons.

12. Implications for event-driven research

1. Separate document-time from transform-time point-in-time validity. Classify every transform before use; treat models trained on later corpora as non-PIT by default.
2. Gate context sources before modelling. Let a layer fail as "untested" rather than fill it with current versions or zeros.

3. Report the effective sample of shared context: distinct decision times and unique events, not asset-time rows.

4. Freeze an a-priori primary specification and a locked evaluation block; treat tuned gains as exploratory until they survive the block.

5. Report nulls with their scope. "This representation did not show incremental value here" is a result that a broader claim must not overwrite, and it is not evidence that no effect exists.

13. Conclusion

Under the frozen point-in-time specification, the surviving public-context feature set did not provide evidence of incremental out-of-time predictive information about Binance full-asset delisting beyond the structured market baseline. The pre-registered verdict is PRE-REGISTERED NULL. This is an absence of evidence of incremental value, not evidence that context has no effect: the point estimates were negative and the development interval included zero. The finding is bounded by what survived the gates. Exchange-specific and asset-specific context could not be reconstructed point-in-time here and remain untested. The comparison could detect only large increments, and its development sign depends on the penalty. Prospective confirmation is pending.

Use of AI tools

The research was carried out with an AI coding agent (Claude, Anthropic) working in the author's repository. The agent wrote and ran the collection, feature, audit and model code for H20.0–H20.3, produced the semantic-audit labels for the H20.1 classifiers (§6), drafted the pre-registration documents and this manuscript, and ran the claim audit that checks every number in the text against the frozen artefacts. The author set the research question and each stage's brief and constraints, authorised each freeze and the single opening of the outcome, decided what to publish, and is responsible for the content. No language model was used to construct any feature (§6).

Data and code availability

The public package is the `contextual-alpha/` directory of the repository that also holds the companion paper (Ramos Silva 2026). It contains:

- the frozen plans, pre-registrations and freeze files with their SHA-256 hashes;
- the collection, build, audit and model scripts, and the tests;
- the evidence tables behind every number, the claim ledger and its audit.

A reader can verify from it that every frozen file matches its recorded hash, and that every number in this paper matches its frozen source. A reader cannot verify from it:

- **the raw corpora**, which are not redistributed; the retrieval scripts document how to obtain them;

- **the inherited H18 inputs** (the event universe and the market bars), which belong to the research repository;
- **the order of events.** The freezes were committed in the author's research repository (H20.0 `a14c347`, H20.1 `e21c264`, H20.2 with Addendum 02 `9932e20`, H20.3 `180def4`), whose history is not public.

The scripts document and audit the procedure; they do not run end to end from the public package alone. Paper tables and figures are regenerated by `paper/scripts/build_paper_assets.py`.

References

Rendered from `paper/references.bib`, the source of truth. Each entry carries the access level recorded for it in `paper/REFERENCE_VERIFICATION.md`; no metadata was inferred.

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Appendix A. Feature-level eligibility (H20.1)

Table A1. H20.1 feature-level eligibility

feature_id	feature_name	layer	family	eligible_h20_2	failed_criteria
F01	c1_fed_release_count_W7	C1	R2	True	
F02	c1_fed_monetary_policy_count_W30	C1	R2	True	
F03	c1_fomc_statement_flag_W7	C1	R1	True	
F04	c1_cpi_release_flag_W7	C1	R1	True	
F05	c1_unrate_release_flag_W7	C1	R1	True	
F06	c1_policy_rate_asof	C1	R4	True	
F07	c1_cpi_yoy_asof	C1	R4	True	
F08	c1_unrate_asof	C1	R4	True	
F09	c2_fedreg_finreg_doc_count_W7	C2	R2	True	
F10	c2_fedreg_crypto_doc_count_W7	C2	R2	True	
F11	c2_fedreg_crypto_doc_count_W30	C2	R2	True	
F12	c2_fedreg_crypto_rulemaking_count_W30	C2	R2	True	
F13	c2_sec_press_count_W7	C2	R2	True	
F14	c2_sec_enforcement_count_W7	C2	R2	True	
F15	c2_sec_crypto_press_count_W30	C2	R2	True	
F16	c2_fed_enforcement_count_W30	C2	R2	True	
F17	c2_cftc_press_count_W30	C2	R2	False	coverage 0.314, 4 years
F18	c2_crypto_regulatory_action_flag_W7	C2	R1	True	
F19	c3_gdelt_crypto_record_count_S7	C3	R2	False	semantic defect K6
F20	c3_gdelt_crypto_record_share_S7	C3	R2	False	semantic defect K6
F21	c3_gdelt_crypto_record_share_en_S7	C3	R2	False	semantic defect K6

feature_id	feature_name	layer	family	eligible_h20_2	failed_criteria
F22	c3_gdelt_crypto_tone_mean_S7	C3	R3	False	semantic defect K6
F23	c3_wiki_crypto_item_count_D7	C3	R2	True	
F24	c3_wiki_crypto_item_flag_D7	C3	R1	False	REDUNDANT with F23 (rho 0.9996)
F25	c3_crypto_stress_flag_D7	C3	R1	False	semantic defect K5; SPARSE_DEGENERATE (nonzero share 0.0)

Appendix B. Fold-level development results

Table A2. Fold-level development results (primary model)

fold	test era	test events	test rows	PR-AUC M0	PR-AUC M2	Δ (M2 − M0)	Δ (M2_core − M0)
1	2022	31	4575	0.3794	0.3675	−0.0119	+0.0003
2	2022	27	4778	0.3608	0.3507	−0.0101	−0.0134
3	2022	27	4830	0.2305	0.2283	−0.0022	−0.0048
4	2022	24	4792	0.1538	0.1423	−0.0115	−0.0124
5	2023	22	4713	0.2034	0.1973	−0.0061	−0.0050
6	2023	20	4666	0.1351	0.1359	+0.0009	+0.0057
7	2023	27	4707	0.1614	0.1689	+0.0074	+0.0064
8	2023	35	4711	0.2017	0.1911	−0.0106	−0.0082

Appendix C. Data sources

These are data surfaces, not literature citations:

- the Federal Reserve press-release feed;
- the Federal Register API;
- SEC newsroom press-release listings;
- CFTC press-release listings (2018–2021 pages only);
- ALFRED real-time data vintages for CPIAUCSL, UNRATE and DFF (Federal Reserve Bank of St. Louis n.d.);
- the English Wikipedia Portal:Current events day pages, read at the revision current at t;
- GDELT 2.0 GKG 15-minute files (H20.1 only; no GDELT feature was eligible) (The GDELT Project 2015);
- Binance spot market data and the frozen H18.1b full-delisting event universe (outcome only).